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-:FOR UPSC IAS ASPIRANTS:-

"AVOID POLITICAL & IRRELEVANT ARTICLES"

Please Try To Read This Completely in 40 Minutes If You

Can't So Then You Have To Increase Your Efforts

All the topics of this UPSC IAS Edition are directly or indirectly important for the prelims & main examination.

There are some topics which can be coded in answer

writing of other topics in the main exam.

No need to worry about US Section 301, we will tackle it... confident of a good trade deal: Goyal

Asks India Inc to get out of ‘comfort zone’, look for opportunities at home, abroad

Express News Service
Mumbai, June 7

DAYS AFTER the Trump administration proposed 12.5% tariffs on India and 53 other countries after a United States Trade Representative (USTR) trade investigation under Section 301 of the US Trade Act of 1974 declared that these countries failed to impose a legal prohibition on “importation of goods produced wholly or in



part with forced labour”, Commerce and Industry Minister Piyush Goyal said Sunday he



Union Minister Piyush Goyal in Mumbai, Sunday. SANKHADEEP BANERJEE



didn’t think there was any need to worry about Section 301, and that “we will tackle it”.

The proposed tariffs have been announced while US and Indian negotiators are finalising a bilateral trade agreement.

Speaking at the *Financial Express* India’s Best Banks Awards in Mumbai, Goyal said US Trade Representative Jamie-son Greer will visit India in the next two weeks for further discussions on the trade deal. Earlier, he had indicated an agreement could be reached by July.

“So this (Section 301 investigation) is really a mechanism

»CONTINUED ON PAGE 2

SECOND PRICE REVISION SINCE CRISIS IN WEST ASIA

LPG cylinder price up Rs 29; OMCs still losing Rs 700: Govt

Ministry: Price of household cylinder lower than that in neighbouring countries

Sukalp Sharma
New Delhi, June 7

THE PRICE of a 14.2-kg domestic LPG cylinder was raised by Rs 29 effective Sunday, the second such price revision since the start of the war in West Asia which triggered a supply crisis.

The increase follows a Rs 60-per-cylinder hike on March 7, taking the cumulative hike to Rs 89 per 14.2-kg domestic cylinder. It will now cost Rs 942 in Delhi — corresponding changes in price have been effected in other parts of the country.

Beneficiaries of the Pradhan Mantri Ujjwala Yojana (PMUY) will continue to pay an effective Rs 642 per cylinder after receiving a subsidy of Rs 300 per refill on the first four refills annually, down from 9 refills announced last year.

Despite the hike, public sector oil marketing companies (OMCs) continue to absorb under-recoveries of about Rs 700 per 14.2-kg cylinder as the market-linked price of the fuel is over Rs 1,600 per cylinder, the Petro-



LPG cylinders being delivered in Nadia, West Bengal, Sunday. PTI

E. EXPLAINED

Consumption down

India’s LPG consumption stood at about 90,000 tonnes a day before the West Asia war. But with the disruption in flows through the Strait of Hormuz and seasonal dip in demand, LPG consumption has declined to about 72,000 tonnes a day, according to recent estimates.

leum Ministry said Sunday.

It also said that the price of domestic LPG in India continues to be lower than in neighbouring countries as the government and the OMCs have been shielding house

»CONTINUED ON PAGE 2

• SHOWS RARE CONTINUITY OF HUMAN ACTIVITY LINKING EARLY TOOL-MAKING TRADITIONS WITH LATER RITUAL, SYMBOLIC AND SOCIAL LIFE

Rock marks, game boards: Wildlife trail leads to prehistoric past in Aravalli

Sophiya Mathew & Abhimanyu Hazarika
New Delhi, June 7

FOUR MONTHS ago, a wildlife trail through the rocky outcrops of the Bhondsi stretch of the Aravalli forests in Gurgaon led wildlife enthusiasts to a stone surface marked with dozens of carefully carved cup-shaped depressions, geometric grids, and a large foot-shaped engraving.

That spot has since been identified as a cluster of ancient petroglyphs (images or designs

carved or engraved on rock), possible stone game boards, and stone tools — part of a wider archaeological pattern emerging from the Aravalli belt where, researchers say, evidence from multiple prehistoric phases appears to survive within the same landscape.

“A chronological continuity is being found here,” said Akash Gupta, a PhD scholar and faculty member at Delhi University’s Satyawati College who has examined the Bhondsi material. “The discovery is significant because it

shows a rare continuity of human activity linking early tool-making traditions with later ritual, symbolic, and social life across this region.”

Gupta said scholars such as French archaeologist André Leroi-Gourhan viewed hand engravings on rock as evidence of a cognitive shift. Placing the latest discovery in context, he said the region has yielded signs ranging from the Lower Palaeolithic to the Mesolithic period.

“Handaxes and cleavers point to early stone-tool traditions, while the tools indicate later Middle Palaeolithic activ-



(Left) Foot engraving on quartzite rock; an Acheulean bifacial hand axe found close to rock art cliff. COURTESY: TEJVEER MAVI, SHAILESH BAISLA AND YATIN VERMA

ity (300,000 to 50,000 years ago). The cupules (cup-shaped depressions) and rock-art-like markings are more likely to belong to later symbolic or ritual phases, especially the Mesolithic, when prehistoric communities are known to have left behind paintings, burial practices and other markers of social life,” he said.

Officials said excavation teams from the Chandigarh Circle of the Archaeological Survey of India (ASI) and the Haryana Archaeology Department have been documenting the site, including the petro-

glyphs and other remains, and have submitted a report.

Kamei Athoilu Kabui, Superintending Archaeologist with ASI’s Chandigarh Circle, said, “The site needs to be protected from the land mafia and further steps in this regard need to be taken by the state government department.”

She said the ASI’s local exploration team, working with local villagers, had found in situ petroglyphs and stone tools.

Anil Tiwari, Deputy Superintending Archaeologist, ASI Chandigarh, said: “A brief report was prepared but there is

still a lot of scope for further work and research.”

According to Shalaish Baisla, a PhD scholar and researcher at the School of Heritage Research and Management, Dr B R Ambedkar University Delhi, the Bhondsi documentation began after wildlife photographers Yatin Verma and Ramkumar shared images of the cup-marked rocks they had come across in the forest in February.

Verma said, “During a bird-watching trip with my team, we heard a leopard calling from a cliff above us. The cliff is a

»CONTINUED ON PAGE 2

[FROM PAGE 1 | FULL REPORTS ON WWW.INDIANEXPRESS.COM]

Domestic LPG price up

holds from spiralling international prices amid the West Asia crisis.

The latest LPG price hike follows increases in the price of other fuels like petrol, diesel, and compressed natural gas (CNG) in recent weeks. The OMCs – Indian Oil, Bharat Petroleum, and Hindustan Petroleum – are grappling with financial strain as they have been selling consumer fuels below market price. Like in the case of domestic LPG, the companies continue to incur losses on the sales of petrol and diesel, despite the recent price hikes.

“The prices of petroleum products in India are linked to the corresponding prices in the international market. The Government, however, continues to modulate the effective price to the consumer for domestic LPG. Any household can buy as many cylinders as it needs at Rs 942. A PMUY (Pradhan Mantri Ujjwala Yojana) beneficiary will additionally receive the direct benefit transfer of Rs 300 a cyl-

inder on the first four refills each year (down from nine a year)... and so pays an effective Rs 642 on those refills; this support is unchanged. Even a non-PMUY household would pay about Rs 700 below the market-linked cost of the cylinder,” the Petroleum Ministry said.

There are over 33 crore households with domestic LPG connections, including 10.6 crore households that avail under the PMUY, which provides an additional subsidy of Rs 300 per cylinder.

The international LPG price benchmark, Saudi Contract Price (CP), has risen sharply through the West Asia crisis. It has gone up to \$790 per tonne in June from \$542.50 in February, marking an increase of about 46%.

India depends on imports to meet about 60% of its LPG consumption, and 90% of those imports came via the Strait of Hormuz, where maritime traffic has been effectively halted since early March. The strait’s closure

has led to a surge in international oil and fuel prices as the waterway accounted for a fifth of global crude oil and liquefied natural gas (LNG) flows.

“The commercial cylinder used by hotels and businesses is revised automatically every month, because its price is a direct pass-through of the international benchmark. The domestic cooking cylinder is not. India used to import 60 per cent of its LPG requirements, and the landed cost of that import tracks the Saudi Contract Price (CP) that Saudi Aramco sets at the start of each month. This is an external price over which the Indian consumer has no control,” the Petroleum Ministry said.

Under-recovery is the gap between the international price of LPG and the regulated retail price in the country. It is absorbed by the OMCs and compensated partly by the government. The cumulative under-recovery on LPG in 2025-26 (FY26) was Rs 60,000 crore, up from Rs 41,338 crore in FY25. In August of last year, the government approved Rs 30,000 crore in budgetary support to the OMCs to partly offset their

losses on domestic LPG sales.

“The under-recovery now absorbed on each domestic cylinder is about Rs 700. The scale of this is visible in the fully market-priced commercial cylinder: the 19 kg cylinder used by hotels and restaurants sells in Delhi at Rs 3,113.50, about Rs 164 a kg, after five increases during the West Asia crisis. The domestic household, by contrast, pays about Rs 66 a kg after the revision. Commercial gas carries a higher rate of tax and larger margins, so it sits above the household’s cost-reflective level; even so, the import-linked cost of a domestic cylinder works out to over Rs 1,600,” it said.

According to the Ministry, domestic LPG prices in India are lower when compared to neighbouring countries like Pakistan (Rs 1,046 per cylinder), Nepal (Rs 1,207), Bangladesh (around Rs 1,225), and Sri Lanka (Rs 1,241). In the developed world, LPG prices have shot up even further – for instance, about Rs 1,755 in the US, around Rs 1,765 in Australia, and about Rs 2,411 in Canada – as per the government.

Goyal

being created, given their (the US) constraints that the Congress is not going to support any of their actions (on reciprocal tariffs)... They are trying to create a competitive edge for India. So I don’t think (we need to) worry about Section 301, we’ll tackle it, it’s our responsibility,” he said.

“The (Section 301) investigation is directed at a particular country,” Goyal said, without naming China.

He said India was keenly awaiting what kind of tariffs would ultimately be imposed under Section 301. “We will protect India’s interests and I am very confident that the (India-US trade) deal will come through. It will be a good deal,” he said.

He, however, reiterated that “India stands for fair and equitable trade” and any deal being signed would be “balanced,” and that the country won’t negotiate with deadlines set.

Goyal noted that in a world where geopolitics is in a flux, “re-alignment of friends and friendships” would be inevitable.

Replying to a question on whether investments from China could be facilitated further given India’s growing trade deficit with that country and the critical inputs required for new-age manufacturing, he

said the government “has no problem with investments from China so long as they are in the desirable sectors and not meant for opportunistic takeover of Indian assets.”

He, however, ruled out any revisiting of India’s decision to stay out of the China-led Regional Comprehensive Economic Partnership (RCEP), saying it was incomprehensible in the first place why the then UPA government joined the talks for the 15-country trade bloc, while the country already had preferential trade pacts with most of these countries, except China.

He said the bulging merchandise trade deficit with China — \$112.4 billion in 2025-2026 — could be progressively reduced by a multi-pronged strategy, including creating competitive domestic capacities wherever feasible, and cracking down on unfair trade practices like dumping and predatory pricing.

He hinted that the UK’s insistence on proceeding with steel import tariff measures after all processes for implementing the India-UK Comprehensive Economic and Trade Agreement (CETA) were almost concluded would lead to India clawing back some concessions it had given in the pact. “On steel, we will have to rebalance with some products which will possibly even hurt the UK businesses. But that’s the nature of trade,” Goyal said. He

spoke of “two other issues” that had cropped up closer to the implementation of the bilateral CETA, and one of these was settled in India’s favour in recent Delhi talks.

On March 19, the UK government announced that it will limit tariff-free steel imports from July 1, 2026, reducing overall quota volumes by 60% compared to the steel safeguard measure. Any imports above these levels will subsequently face a 50% tariff.

Goyal ruled out putting any curbs on investors repatriating their investments from India or outbound capital, citing the country’s comfortable foreign exchange reserves and Current Account Deficit (CAD) remaining well under 2%. “Our financial services are largely about India’s best banks and financial services. We’ve been too inward-looking. We have been content with a large Indian market. We have to shake our conscience, shake ourselves to start exploring world markets,” he said.

Goyal called upon Corporate India “to get out of its comfort zone” and look for investment and business opportunities both in India and abroad. “We have to get out of our comfort zone in every area. I mean, for that matter, whether it’s banking, financial services, taking UPI internationally, publishing, jewelry, construction, education, culture, even sports,” he said.

Aravalli

popular spot for yoga and recreational activities among local residents. After the calls ceased, we hiked to the top to inspect the area. There, I noticed several engravings that resembled the petroglyphs I had seen in reports and news articles.”

Baisla later visited the site with Tejveer Mavi, “a local expert from Kot village whose field knowledge of the Aravalli landscape has helped researchers identify several prehistoric markers across the ridge”.

The Bhondsi cluster, located near the Bhuvneshwari Mata Mandir temple complex in the belt running from Delhi’s ridge through Gurgaon and Faridabad, is unusual because of the concentration and variety of markings on the exposed rock surfaces.

The team has documented cupules, geometric engravings, possible two-player game-board patterns and a large foot-shaped engraving, said Baisla, adding that this may have had ritualistic or ceremonial significance.

The team has been using LiDAR (Light Detection and Ranging) scanning – a laser-based remote-sensing technique that captures the shape and depth of surfaces in high detail – to create detailed digital records of the petroglyphs.

Delhi’s infant feeding practices see sharp dip across nutrition indicators: NFHS

Early initiation of breastfeeding is widely regarded as critical for newborn survival

Ankita Upadhyay
New Delhi, June 7

CHILD FEEDING practices in Delhi have significantly deteriorated over the past three years, according to new data released by the National Family Health Survey (NFHS-6) 2023-24, with dips observed across nearly every key indicator of infant and young child nutrition.

The share of babies breastfed within one hour of birth dropped from 51.2% in NFHS-5 (2019–21) to 45.1% in the latest survey. The drop is significant: early initiation of breastfeeding is widely regarded as critical for newborn survival, helping infants receive colostrum – the antibody-rich first milk – and establishing feeding patterns linked to sustained breastfeeding in early life.

The data shows that exclusive breastfeeding among infants aged below six months saw an even sharper decline: from 64.3% to 48.3%. Exclusive breastfeeding is defined as feeding only breast milk – including expressed milk or milk from a wet nurse – with no other food or drink, not even water, except oral rehydration solution and prescribed drops or syrups, such as vitamins or medicines.

The World Health Organization recommends exclusive breastfeeding for the first six months of life, noting that breastmilk alone provides all required energy and nutrients while protecting against common childhood illnesses.

Overall breastfeeding coverage among infants under six months also slipped, with the proportion of babies receiving breastmilk declining from 96% to 88%. The share of infants who were exclusively or pre-

Feeding practices, nutrition received by children in Delhi

	Urban (%)	Rural (%)	Total as per NFHS-6 (%)	NFHS-5 (2019–21) (%)
Children under 3 yrs breastfed within 1 hr of birth	45.1	46.6	45.1	51.2
Children under 6 months exclusively breastfed	47.5	*	48.3	64.3
Children under 6 months currently breastfed	88	*	88	96
Children under 6 months exclusively or predominantly breastfed	71.9	*	72.2	83.9
Children aged 6–8 months receiving breastmilk and complementary food	52.2	*	52.5	62.9
Children aged 6–23 months receiving breastmilk and adequate diet	11	10.4	10.9	18
Children aged 6–23 months receiving adequate diet, but not breastmilk	12	*	11.9	8.6
Children aged 6–23 months receiving adequate diet	11.3	8.9	11.2	16
Stunted (height-for-age)	26.4	28.5	26.4	30.9
Wasted (weight-for-height)	15.1	10.3	15.0	11.2
Severely wasted (weight-for-height)	4.2	0.0	4.1	4.9
Underweight (weight-for-age)	27.4	26.6	27.4	21.8
Overweight (weight-for-height)	1.8	0.3	1.7	4.0

**16%**

Decline in exclusive breastfeeding (under 6 months)

**7.1%**

Decline in children getting breastmilk and an adequate diet (6-23 months)

**5.6%**

Underweight children

dominantly breastfed fell from 83.9% to 72.2%.

Among older infants, feeding practices also deteriorated. The proportion of children aged 6–8 months receiving complementary food alongside breastmilk declined from 62.9% to 52.5%. Only 11.2% children aged 6–23 months were receiving a minimum adequate diet, down from 16% – suggesting that nearly nine in ten children in this age group are not receiving basic standards of dietary diversity and meal frequency.

Doctors and public health experts say the trend points to a broader breakdown in early childhood nutrition practices. Dr Uma Vaidyanathan, director of obstetrics and gynaecology at Fortis Hospital Shalimar Bagh, termed the decline “a matter of concern”, noting that despite

routine counselling, many mothers are unable to continue exclusive breastfeeding for the recommended duration.

She said returning to work early after childbirth is a major factor, with many women introducing formula or other feeds within a few months to accommodate care-giving arrangements. The issue, she added, is not limited to working mothers. “Even among homemakers, we have observed a growing tendency to discontinue breastfeeding early, often due to lack of sustained encouragement,” Vaidyanathan said.

“There is generally no such thing as a biological failure of lactation,” she said, adding that most women are physically capable of breastfeeding for one to two years, but increasingly choose not to continue.

Dr Neelam Benara, Consultant, Obstetrics and Gynaecology at Kailash Hospital, Noida, said the NFHS-6 findings point to a worrying decline across multiple indicators. The reductions in early initiation, exclusive breastfeeding, timely complementary feeding and dietary adequacy, she said, suggest that many children are not receiving optimal nutrition during a critical window of growth and development. While the survey does not identify specific causes, she said possible barriers include gaps in breastfeeding support, caregiver awareness and nutrition counselling.

Dr Vaidyanathan cautioned against a growing reliance on formula feeding, saying it may be associated with higher rates of digestive issues and allergy-like symptoms.

DELHI BIRD ATLAS CITES PRESENCE OF ARAVALLIS, FLOODPLAINS, AND A MIGRATORY FLYWAY AMONG REASONS

‘Among global capitals, Delhi ranks second in bird diversity’

Sophiya Mathew
New Delhi, June 7

THE DELHI Bird Atlas, released on June 5, which claims to have mapped the distribution and abundance of bird species across the city for the first time, has placed it “second only to Nairobi among the world’s national capitals in bird diversity”.

The atlas cites the rare mix of the city’s geography, including the northern edge of the Aravallis, proximity to the western Himalayas, the Yamuna and Sahibi floodplains, and its location near the Central Asian Flyway (CAF) — a key bird migration route stretching from the Arctic down to the Indian Ocean — as the primary reason behind the bird diversity.

The atlas was developed by the Delhi Forest Department, Bird Count India, World Wide Fund for Nature (WWF)-India and other partner organisations, with the participation of birdwatchers, students, researchers, forest staff and

citizen volunteers.

According to the atlas, “The Delhi Bird List now stands at 471 species”, excluding another 22 species that have not been recorded since 1975.

The atlas covers 11% of Delhi’s area in its sampling design. The survey started in January 2025, takes place twice a year, in winter and summer, and will run for at least two years. Winter surveys document resident species and winter migrants, while summer surveys record breeding activity, resident birds, summer migrants and early-returning migrants.

In the first year, 221 species were recorded across the city, including 200 in winter and 152 in summer. Of these, 126 were resident species, 81 winter migrants and 14 summer migrants. Nearly half the recorded bird assemblage comprised invertebrate feeders, that is, 108 species or 48.87%. Plant and seed feeders accounted for 37 species, omnivores for 34, vertebrate and carrion feeders for 33, and fruit and



The Ultramarine Flycatcher and (right) the River Lapwing were among birds surveyed. SURVEYOR MOHIT MEHTA

nectar feeders for nine.

The atlas also records 18 endemic bird species, that is, species found exclusively in one specific geographic region. It also lists several vulnerable bird species, including the Endangered Egyptian Vulture and Black-bellied Tern, the Vulnerable River Tern and Common Pochard, and Near Threatened species such as the Black-tailed Godwit, Ferruginous Duck,



Painted Stork, Black-headed Ibis and Oriental Darter.

Unlike other bird lists, the atlas used a grid-based method. Delhi was divided into grids of 6.6 km by 6.6 km, subdivided into smaller quadrants and subcells. A total of 145 subcells were randomly selected for sampling to reduce bias and ensure representation across habitats. Each subcell required four 15-minute checklists, with birds recorded

on foot and uploaded to eBird. The atlas says this “systematic approach” of collecting “equal-effort lists from randomly placed grids across the city” helps reduce sampling bias and ensures that results reflect the “true distribution and abundance of species”.

Experts associated with the atlas attribute Delhi’s bird diversity to the city’s position at an ecological intersection: the Ridge brings the Aravalli landscape into the Capital, while the Yamuna floodplains and associated wetlands provide refuge for waterbirds and migrants. The city’s proximity to the western Himalayas, meanwhile, facilitates seasonal movement. The atlas notes that Delhi “lies almost at the heart of the Central Asian Flyway”, making it “a seasonal haven for species arriving from as far as Central Asia and the Arctic”.

The exercise, according to the atlas, provides “a robust scientific baseline” against which future atlases, to be con-

ducted five years from now and beyond, can measure ecological change. Repeated surveys will help track shifts in bird populations, habitat preferences and environmental stress over time, officials said.

Meanwhile, the atlas also describes Delhi as “a meeting point of history, geography, and biodiversity”, noting that such diversity persists despite the pressures of one of the world’s most populous urban regions. It underlines the importance of protecting “green and blue spaces that sustain it”, pointing to forests, wetlands, riverine systems, grasslands, urban greens and floodplains as habitats that continue to support birdlife. The Yamuna floodplains, in particular, are identified as a key ecosystem for conservation and habitat restoration.

The Delhi Bird Atlas was prepared by Dr Lynette Gomes, nodal officer, Delhi Forest Department; Pankaj Gupta, state coordinator, Bird Count India; and Arnab Gupta.

JAISHANKAR, SUGIONO HOLD TALKS

India, Indonesia deepen cooperation in defence, maritime security, infra

PM Modi’s visit to Jakarta also discussed

Shubhajit Roy
New Delhi, June 7

EXTERNAL AFFAIRS Minister S Jaishankar and his Indonesia counterpart Sugiono held bilateral talks on Sunday as the two nations agreed to advance co-operation in defence, maritime security, digital connectivity and infrastructure. The two leaders also discussed Prime Minister Narendra Modi’s likely visit to Jakarta next month.

Jaishankar co-chaired the 8th India-Indonesia Joint Commission Meeting with Sugiono in New Delhi.

After the bilateral meeting, Jaishankar posted on X, “... Our Comprehensive Strategic Part-



External Affairs Minister S Jaishankar holds talks with his Indonesian counterpart Sugiono in New Delhi, Sunday. PTI

nership has witnessed a strong growth in recent years. Held substantive discussions on our political, defence & security, maritime & shipping, trade, fintech, health, pharmaceuticals, fertiliser, critical mineral as well as tourism, education and cultural cooperation.”

He also said he “exchanged perspectives on regional develop-

lateral coordination and deepening India — ASEAN ties”.

Sugiono said, “... We discussed key priorities under the Indonesia-India Comprehensive Strategic Partnership, from trade and maritime security to digital connectivity, infrastructure, health and people-to-people ties.”

“Looking forward to Prime Minister Narendra Modi’s upcoming visit to Jakarta,” said

E. EXPLAINED

Historical links

From struggle against colonial powers, democratic traditions and pluralist culture, India & Indonesia share deep linkages. Independent India and Indonesia became chief votaries of independence of Asian and African countries, the spirit of which led to Bandung Conference of 1955 and later formation of NAM in 1961. Indonesia is a key pillar of India’s Act East Policy. The PM’s likely visit will give a fillip to new sectors like defence & semiconductors.

Sugiono, adding that it will be an important opportunity to deepen cooperation between both nations.

RULES OUT THIRD-PARTY MEDIATION

Want to resolve border issues with India via diplomacy, says Nepal Foreign Minister

Divya A
New Delhi, June 7

NEPAL WANTS to resolve the border row with India through diplomacy as no problem is too large if both sides sit across with an “open heart”, Nepal Foreign Minister Shishir Khanal said in New Delhi on Sunday, talking to reporters just before concluding his two-day official visit to India.

Khanal said the new government in Kathmandu refuses to look at India through the “distorted, hyper-sensitive lens of 21st century geopolitics” and instead, wants to build a mutually beneficial relationship for overall prosperity of both nations. This came a day after Khanal held wide-ranging talks with External Affairs Minister S Jaishankar, which included discussions over the boundary issue.

Khanal's remarks comes a week after Nepal Prime Minister Balendra Shah's remarks on the India-Nepal boundary row. Speaking in the Nepal Parliament on May 31, Shah had stated that the border dispute with India is “not one-sided”. He claimed that after taking office, he discovered Nepal had also encroached on Indian territory in several places, just as India had done to Nepal.

Later that day, Nepal's Foreign Ministry clarified that the cross-border encroachments were primarily technical issues related to the vaguely defined “no-man's land” along already demarcated sections of the open border.

Playing down those remarks,



Nepal Foreign Minister Shishir Khanal addresses the media in New Delhi on Sunday. ANI

Khanal said on Sunday that the priority of the new Nepal government is fast economic growth and it is with that lens that they want to upgrade ties with India, acknowledging at the same time that there are “out-standing issues, including border and boundary”.

“There are active mechanisms that exist between Nepal and India. Field survey team is actually on the border, working together. We agreed during discussions with Jaishankar to activate all these mechanisms to solve border issues,” he said.

No problem is too large and no boundary too complex when we sit down with an open heart, Khanal said, adding, “We look at India with an open heart, clear eyes, and with a single, transparent agenda: the economic transformation of Nepal.”

“Rather than engaging in hyper-nationalistic grandstanding, we are pursuing a calm, data-driven approach to resolve issues,” he said.

FULL REPORT ON
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ARTIFICIAL INTELLIGENCE

Is a text AI-aided? Science, limits of detection tools



ABHINAV CHAKRABORTY

AS ARTIFICIAL Intelligence (AI) writing aids become more popular, authors and writers are navigating a tricky territory — there is almost no way to conclusively prove that they did not use AI in any stage of the writing.

The controversy around AI authorship exploded recently when the British literary magazine *Granta* announced the regional winners of its annual Commonwealth Short Story Prize, and three of the five winners faced accusations of AI use.

What makes the debate murky is that the AI detection tools generally used are far from fool-proof. In case of *Granta* winner Jamir Nazir, AI detector tool Pangram was used to term his story *‘The Serpent in the Grove’* (winner for the Caribbean) “100% AI generated”.

What is machine learning?

To understand how these tools claiming to detect AI-generated text work, one must first know the science of machine learning (ML). In simple terms, ML refers to the use of data and statistics to build an AI system: this is done by feeding large data-sets into a computer so that it can recognise patterns.

“So you would take lots of examples of AI-written content and human-written content and feed it to a big model to do the classification for you. The model, through data, learns signals like, ‘Oh, AI models tend to use em dashes’, or use words like ‘imperative’ or ‘delve’. These are statistical patterns that large ML models can learn when they’re fed lots of examples of both human writing and AI writing,” Danish Pruthi, assistant professor at the Indian Institute of Science Bengaluru, told *The Indian Express*.

There are also some AI ‘tells’. Pruthi said that besides em dashes or certain words, other tells include text that is organised in bullet points, often with a heading. Also, AI-generated text tends to conclude things neatly, with Pruthi saying that “human conclusions sometimes introduce new content, but model conclusions rarely do”.

He also cited the instance of “negative parallelism”: a rhetorical writing style

TEACHING, LEARNING AI

MACHINE LEARNING AND AI DETECTION

- Machine learning involves use of data and statistics to build an AI system.
- Done by feeding large datasets into a computer system so that it can reason at human or even superhuman levels.
- Requires lots of examples of both AI-

and human-written content to be fed into the system.

- The system learns to identify statistical patterns in form of words, phrases, and punctuation, thus classifying content as AI-written or human-written.

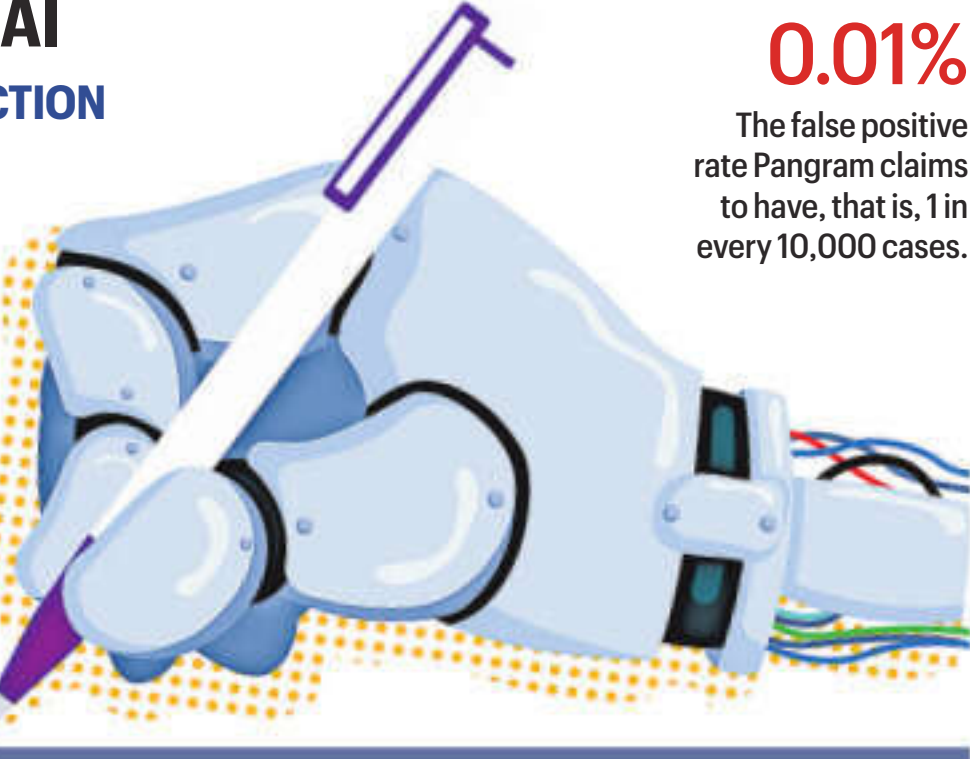
AI ‘TELLS’

- Refers to the signs that a piece of text is AI-generated.

- Excessive use of em dashes, or words not used generally in conversations.

- Text organised in bullet points, often with a heading of what the bullets are about.

- Negative parallelism, a rhetorical writing style marked by the formulaic “Not X, but Y” structure.



0.01%

The false positive rate Pangram claims to have, that is, 1 in every 10,000 cases.

marked by the formulaic “Not X, but Y” structure. “For instance, ‘These headphones are not just hearing devices, but sound-cancelling devices.’ Models are very commonly doing that now,” he said.

As to where these tells come from, Pruthi said research was ongoing but there were no clear answers yet.

“One common hypothesis is that after you pre-train a model, you post-train it to make it safe and useful and able to follow instructions. That’s typically done by contracting annotators and data vendors who create examples to answer questions of different types,” he said.

“A lot of those datasets, which are private and constructed by large frontier labs, have these cues. People who are writing those answers write in this way, and therefore models replicate that behaviour,” he added.

Can AI chatbots be good detectors?

Pruthi said people now want to move beyond this “binary framing” of what is AI versus human. “They are figuring out what the extent of collaboration is. Is it lightly assisted by an AI, moderately assisted by an AI, [or] heavily assisted by an AI?”

According to Pruthi, asking Claude, ChatGPT, or Gemini whether something is AI-written is a “very bad idea”. “The model is not specifically trained for this. So it might take an educated guess, but that isn’t going

Need for openness

Friedman said onus was on everyone in publishing or related fields to be on the same page, despite “very good reasons to be anti-AI”.

How reliable are these tools?

Pruthi said that Pangram — which claims a false positive rate of 1 in every 10,000 cases (0.01%) — is quite reliable, which was backed by some independent studies.

But he sounded a word of caution, saying that an ML model would “obviously not be 100% accurate all the time”. Using the analogy of spam classification in emails, he said: “We still develop ML models to detect what is spam and what is not. There too, the content, the words used, the way it is phrased — all that is helpful to figure out whether it is spam or not. We still get a few

to be very accurate... This happens to be a task where you care about accuracy a great deal because of it being high stakes,” he said.

The other key difference, Pruthi said, is that a lot of detectors are tuned in such a way to ensure “false positives are low”. A false positive refers to an instance when a detector flags something human-written as AI-generated, as opposed to a false negative, wherein an AI-generated text might pass off as human.

AI detectors are also distinct from tools detecting plagiarism. Pruthi said since plagiarism largely concerns copying intellectual work without attribution, “plagiarism detectors tend to sophisticatedly score how well a given idea/work matches with existing work”. “On the contrary, AI detectors just try to estimate whether a piece of text could be AI-generated given lots of examples they’ve seen,” he said.

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examples wrong.”

This is because these tools have limitations. According to Pruthi, ML models are more likely to be wrong when there are fewer words, since there aren’t enough signs to tell confidently whether something’s AI-written or human-authored.

Another limitation is what Pruthi called “low-entropy text”, which refers to text that is generally precise and accurate in nature, and thus hard to classify.

“If I ask you, ‘Give me all the states of India in alphabetical order.’ This has one clear, definitive answer, whether from a model or a human,” he said.

Similarly, code — written instructions that tell a computer to execute a certain task — can also “be tricky to detect” at times. “There’s only a certain way it can be written,” Pruthi added.

Pruthi mentioned another limitation — one that he and his colleagues presented in a recent paper at an ML conference — in a scenario when one slightly polishes a piece of text using a language model. “Even though the base ideas and content were written by you, instead of saying that it is slightly edited or mixed text, models might flag it as fully AI-generated,” he said.

This can dissuade writers from even refining their writing using AI, out of fear that their work could be mistakenly flagged as AI-generated. “There’s a lot of AI slop on the internet right now. A lot of Kindle books that are published are fully AI-written,” he said. “So if a good detector is able to weed out most of that and at least label that this is AI-generated, then maybe a careful reader can choose not to consume that content. In some way, good AI detectors are helping channel attention to legitimately human-written content.”

How should writers, publishers navigate this?

Recently, Nobel Prize-winning Polish writer Olga Tokarczuk’s comments about using AI for research while writing her last novel had invited criticism. Tokarczuk later clarified she did not use AI in the writing process itself.

Jane Friedman, an American publishing professional with over two decades of industry experience, told *The Indian Express*, “Everyone needs to be on the same page about where these tools are touching the process and, to the best of everyone’s ability, track how they’re being used.”

“One of the problems here is that everyone is kind of doing their own thing behind the curtain. Part of that has to do with the taboo around it and everyone being uncertain about the technology and the different attitudes towards it,” she added.

SHIPPING

Why energy tankers are using risky dark fleet ploy to cross Hormuz

Sukalp Sharma

New Delhi, June 7

FOR YEARS now, ships involved in sanctioned and illicit trade have “gone dark” to avoid detection — that is, switched off their automatic identification system (AIS) transponders to make themselves virtually invisible to authorities and other vessels.

This dark shipping practice has largely been limited to the “shadow fleets” transporting sanctioned Iranian and Russian oil. But as the US-Iran war drags on, for energy tankers looking to sail through the blocked Strait of Hormuz, avoiding detection is increasingly becoming a necessity. This has brought this risky tactic out of the margins.

Surge in dark transits

According to International Maritime Organization (IMO) guidelines, AIS should always be in operation when a ship is sailing or at anchor, unless the master of the

ship decides that broadcasting the information is a threat to its safety or security.

Switching the transponder off makes the vessel untrackable and invisible to other ships’ systems, which can significantly increase the risk of collisions at sea. This risk becomes even greater in crowded waterways such as the Strait of Hormuz.

According to energy and freight markets analytics firm Vortexa, dark transits through the Strait of Hormuz — between March 1 and late May — represent 57% of all transits recorded over the period.

This trend of ships going dark while transiting the strait has intensified in recent weeks. The share of outbound laden vessels transiting dark stood at 58.5% in March, slightly dipped to 54% in April when vessel movements through the waterway picked up slightly through corridors regulated by Iran, and then rose sharply to 65.2% in May, as per Vortexa.

“Commercial operators are simulta-

India-bound ships too

- India depends heavily on the Strait of Hormuz, with over 40% of its oil imports, 60% of its LNG imports, and 90% of its LPG imports coming through this narrow passage.

- Many of the energy cargoes that have reached India via the Strait of Hormuz over the past three months went dark as they crossed the strait.

neously adapting both operationally and physically to the environment through prolonged dark operations, altered routing, escort behaviour, and possible vessel hardening measures. Taken together, the Strait of Hormuz is increasingly functioning as a contested low-visibility operating environment where commercial transit continues, but under growing military pressure, degraded transparency, and sustained enforcement risk,” maritime intelligence provider Windward said in a note last month.

Of the energy cargoes being transported by tankers that are going dark to cross the Strait of Hormuz, crude oil and condensates account for around 40%, ‘clean products’ (like petrol, diesel, jet fuel, and naphtha) account for about 25%, ‘dirty products’ (like bunker oil and bitumen) have a share of about 18%, and LPG accounts for about 14%, shows Vortexa data. LNG tankers, hitherto absent from dark shipping activity in the region, appear in

the data from April into May.

Accepted protocol

Ship tracking data shows that tankers belonging to or chartered by the national energy companies of countries such as the UAE, Qatar and Saudi Arabia are also going dark to cross the strait. In March, non-Iranian operators accounted for 37% of dark outbound laden transits through the chokepoint, which rose to 56% in April and 67% in May.

Vortexa analysis shows that UAE-linked vessels account for the largest share of dark transits at around 27%, followed by Iraq at about 11%, and Qatar at around 10%. Saudi Arabia, Kuwait, and Bahrain together account for another 9%.

Experts point out that with global energy flows disrupted due to the crisis, there is increasing pressure on major energy producers from the Gulf to keep exports going to the extent possible.

LEGAL

For survivors of trafficking, SC plan focuses on consent, dignity

Amaal Sheikh

New Delhi, June 7

THE SUPREME Court last month said that considering that the “existing vacuum seriously impairs the fundamental rights” of the victims of trafficking for commercial and sexual exploitation (CSE), it was “left with no option but to pass detailed directions” by framing a Victim Protection Plan. This will operate until Parliament enacts a law on the protection and rehabilitation of victims of CSE.

Background

The litigation traces to a 2004 petition filed by Prajwala, a Hyderabad-based anti-trafficking organisation, which argued that victims of CSE were treated as “criminals” rather than “victims or survivors” and that the absence of a comprehensive “Victim Protection Plan” made rescue and rehabilitation efforts ineffective.

In December 2015, the SC disposed of the petition after noting that the government was committed to establishing an Organised Crime Investigation Agency (OCIA) and constituting an Inter-Ministerial Committee to prepare a comprehensive anti-trafficking law. But the OCIA proposal was dropped, with the Centre instead empowering the National Investigation Agency (NIA) to investigate trafficking offences in 2019. Prajwala returned to the SC in 2022, alleging non-compliance with the commitments on which the 2015 order rested.

Rehabilitation, no criminalisation

On dignity, the SC said that every person possesses dignity “simply because they are human”. Trafficking violates this directly as “the entire transaction proceeds as though their humanity is irrelevant and they are nothing more than mere objects”.

On rehabilitation, it held that victims of trafficking for CSE have a constitutional right to rehabilitation.

The bench examined Section 17 of the Immoral Traffic (Prevention) Act (ITPA), 1956, which governs what happens after a person is removed from a brothel during a raid. It noted that the provision treats women

trafficked into prostitution, a woman who was trafficked but later continued in sex work voluntarily, and a woman who entered sex work voluntarily, alike. A “one-size-fits-all approach” risks becoming “counterproductive” and “may violate their rights and dignity”.

The court held that magistrates must first conduct an inquiry to identify voluntary adult sex workers, for whom the question of “rescue” does not arise. Victims’ consent must be “the primary and governing consideration” in decisions on detention in a protective home or reintegration with family.

Debangana Chatterjee, assistant professor of social science at the National Law School of India University, told *The Indian Express* that the distinction the court draws between voluntary and involuntary sex work does not hold on ground. Consent, in practice, operates under what she described as “dire structural conditions” that make a clean determination of voluntariness difficult. “Unless you have a sense of what exactly ‘voluntary’ entails, this becomes really difficult to decipher.”

Victim Protection Plan

The plan states that victims should not be treated like criminals, and rehabilitation cannot be imposed without “free and informed consent”. The bench directed that anti-human trafficking units be headed by officers of Deputy Superintendent of Police-rank and be notified as police stations for registering and probing trafficking cases.

During rescue operations, victims are not to be arrested, subjected to verbal or physical abuse, or photographed and filmed in a manner that reveals their identity.

Post rescue, victims cannot be kept in lock-ups or detained overnight in police stations, must be provided legal aid, medical care and counselling, and must be produced before the appropriate authority without delay. Before ordering long-term custody or rehabilitation, magistrates are required to hear the victim and, where necessary, conduct an inquiry into whether an adult woman is engaging in sex work voluntarily.

GLOBAL

100 days of Iran war: Behind Tehran linking ceasefire more to Lebanon than Gaza



EXPERT EXPLAINS

BASHIR ALI ABBAS

SENIOR RESEARCH ASSOCIATE, COUNCIL FOR STRATEGIC AND DEFENSE RESEARCH

AS THE US-Iran war touches 100 days, the two countries have entered into a new equation that rests on neither side breaking off talks substantially, while continuing to trade limited strikes at strategic targets. The Strait of Hormuz remains blocked.

Even as this equation represents Iran’s strengthened bargaining position, Tehran faces an urgent need to translate this leverage into tangible gains, not least due to its continuing economic crisis, with year-on-year inflation touching 80%.

Iran’s need for decisive outcomes

For Iran, Washington’s overall preference for relying on its own blockade of the Hormuz without a full-scale war (for which its munitions still run low) is a concern. As several analysts have noted, the more days pass before a decisive outcome for Iran (unfreezing Iranian funds, sanctions removal, an end to the US blockade), the longer the break in Iran’s strategy of maintaining multi-pronged pressure.

Ultimately, Hormuz restrictions are an indirect leverage — they are sufficient to deter full-scale American/Israeli attacks but evidently insufficient to force the US towards decisive concessions.

Hence, on June 5, Advisor to the Supreme Leader, Mohsen Rezaee, declared to *CNN* that the US failing to unfreeze \$24 billion in blocked Iranian assets and lifting the blockade would prompt Iran to expand its punitive actions to the Mediterranean, the Red Sea, and the Bab-el-Mandeb strait. Among Iran’s allies, while the Yemeni Houthis’ ability to threaten the latter two has been decisively proven, Lebanon’s Hezbollah

maintains an underutilised ability to threaten military and commercial activity in the Mediterranean through its anti-ship cruise missiles.

However, while Iran maintains its reliance on UNGA Resolution 3314 to deem US bases in Arab countries as legitimate targets in response to limited US strikes on Iran, it is Israel’s war in Lebanon which potentially allows Tehran to generate new *casus belli*.

Despite a US-brokered ceasefire between Israel and Lebanon (Hezbollah) in April, Israel’s campaign in South Lebanon continued largely unabated, until Iran once again threatened to call off talks over it. By June 3, tripartite negotiations led to a joint statement by Israel, Lebanon, and the US, recommitting all parties to the April ceasefire. US President Donald Trump even said the US was “speaking to Hezbollah” and that he “would like to meet” Iran’s new Supreme Leader Mojtaba Khamenei.

Linkage of US war with Israel’s in Lebanon

Lebanon is not important to Iran solely



An Israeli flag hangs on a destroyed building in southern Lebanon. AP

due to the formidable value that Hezbollah brings to Tehran’s strategic power projection. Presently, the Israel-Lebanon ceasefire has a new characteristic — a fragile agreement that Iran knows Israel will inevitably violate.

This provides Tehran a unique strategic option: to determine that Israel’s attacks in Lebanon warrant a tighter chokehold on Hormuz or a resumption of its missile attacks, at any point when Tehran perceives Washington to be slow-peddalling a deal.

Already by June 4, Iranian Foreign Minister Abbas Araghchi reaffirmed that “this war will end only when it ends in Lebanon as well”. Combined with Rezaee’s statements, it is clear that while protecting Hezbollah is a clear Iranian objective, Lebanon principally serves as a pressure point to force Washington towards concessions vis-à-vis Iran itself.

For perspective, Iran has refrained from officially linking the ceasefire with Israel’s war on Palestine, or at least making it an immediate imperative. Apart from reports from the semi-official *Tasnim News* that Tehran also requires Israel to cease hostilities in Gaza, Iran’s negotiating posture since April has prioritised Lebanon. This is despite Israel’s Gaza campaign pre-dating the current war on Lebanon, causing significantly greater casualty figures, and prompting in-

ternational investigations into Israeli conduct. Iran’s regional posture has also been articulated historically as integrally tied with Palestine (Jerusalem or Al Quds).

Among other things, this reinforces Iran’s awareness that Washington is more likely to pressure Israel on Lebanon than on Palestine; a compromise Iran is willing to make for its immediate strategic needs. The alternative would be a bogged-down Iran with reduced bargaining space in the face of the marginal likelihood of Israel withdrawing from Gaza, even if Tehran threatens military action.

Ultimately, these developments still retain both Iranian and American preference for a deal. However, it is now increasingly evident that just as Washington sought favourable terms through military action in March, Tehran is inching closer to adopting a similar posture. Functionally, this implies that organic pressures are building within the US-Iran equation that will either decisively break the ceasefire, or force Washington and Tehran into at least an interim arrangement, pending a larger deal.

The Editorial Page

MONDAY, JUNE 8, 2026

Amid global shocks, activate growth drivers within domestic control

Facing stress, economy will need a leg up

THE INDIAN economy grew at a healthy pace in the fourth quarter (January-March) of the last financial year (2025-26), despite the conflict in West Asia that had begun to cause widespread disruptions to global energy markets and trade towards the end of the quarter. GDP growth has been pegged at 7.8 per cent for the quarter by the National Statistics Office, driven by a buoyant services sector. This takes growth for the full year to 7.7 per cent — the highest in the last three years. However, at 8.9 per cent, nominal growth was the lowest. It was also significantly lower than the 10.1 per cent assumed in the Union budget.

The disaggregated data shows that the services sector grew at almost 10 per cent in the fourth quarter, with both segments — trade, hotels, transport, and communication and financial, real estate, IT and professional services — growing at a fairly healthy clip. Agriculture continued to hold up, while manufacturing growth slowed down. On the other hand, both private consumption and investment stayed steady. Growth over the full year — a period marked by tremendous uncertainty on the external front owing to Donald Trump's tariff policies — is likely to have been supported by both fiscal and monetary policy. Measures such as the rationalisation of GST rates and low interest rates would have provided support to economic activities.

The outlook for the ongoing financial year (2026-27) is, however, less buoyant. In the June Monetary Policy Committee meeting, the RBI has projected the economy to slow down to 6.6 per cent this year, down from its earlier assessment of 6.9 per cent. This implies that growth this year is likely to be around 1.1 percentage points lower than last year. Some estimates are more pessimistic. Higher energy and commodity prices and continued supply chain disruptions, along with the possibility of a weak monsoon season, are heavily weighing down expectations. Economic momentum is expected to be particularly hit during the first half of the year. Growth is expected to slow down, while inflation is likely to edge upwards. Much will depend on how long the conflict in West Asia drags on. Then there is the stress on the balance of payments and the currency to contend with. On their part, the Union government and the RBI have taken several steps to shield the economy during this period of stress. Last week, measures were announced to attract foreign capital and ease the pressure on the rupee. But, considering the severity of the situation, more steps may be needed.

Rain & fertiliser shortfall can be an opportunity

INDIA'S FARM sector grew by 4.2 per cent in 2024-25 and 3.2 per cent in 2025-26 on the back of two good monsoon years. Rainfall for the country as a whole was 104 per cent of its long period average (LPA) in 2024 and 110 per cent in 2025, with the southwest monsoon season (June-September) alone registering nearly 8 per cent surplus precipitation in both years. This year, the prognosis isn't great due to an El Niño, expected to return after a strong event from around June 2023 to April 2024. Global models indicate a weak-to-moderate El Niño developing in the current monsoon season. The India Meteorological Department has already forecast a below-normal monsoon, with overall rainfall at 90 per cent of LPA for the four-month season.

But the hit to agriculture may not come in the kharif crop season, whose plantings have just about commenced and will pick up as the monsoon advances. The real impact might be during the rabi (winter-spring) season, with strong-to-very strong El Niño conditions projected to emerge through November-January. El Niño is known to not only suppress the monsoon rainfall but also lead to warmer and shorter winters in India. If its intensity is relatively mild during the monsoon months and peaks after November, the rabi crops are the ones that will bear the brunt. The same goes for fertilisers, where the stocks seem reasonably comfortable to meet the requirements of the kharif season. The effects of the supply shock from the ongoing West Asia conflict and blockade of the Strait of Hormuz may be felt only in rabi. This applies even to food inflation, which has not spiralled so far (unlike energy, metals or industrial raw materials), thanks to record 2025-26 crop production levels both in India and globally.

India should turn the potential double whammy of rain and fertiliser shortfalls into an opportunity. The country imported an all-time-high 16.9 million tonnes (mt) of vegetable oils in 2025-26 and 7.3 mt of pulses the previous fiscal. The Narendra Modi government must work towards making 2026 India's year of millets, pulses and oilseeds, which also consume less water and fertilisers — especially urea (nitrogen) — than rice, wheat or sugarcane. Indian farmers respond well to price signals. They will grow these crops once assured of minimum support prices (MSP), either through direct government procurement or payment of difference over open market rates. Such MSP assurance should, however, be conditional upon cost/market-driven pricing of water and fertilisers, which will promote efficient use of these scarce resources.

THE GROWTH-inflation mix is set to worsen for India this fiscal year, largely due to the West Asia conflict and expectations of a sub-par monsoon — factors not in our control. The longer the conflict persists, the greater the risk to growth and upside to inflation.

To be sure, GDP growth was unscathed in 2025-2026, according to the latest estimates of the National Statistical Office (NSO), despite the tariff turbulence of 2025 and the conflict, which began towards the end of February this year.

The NSO now sees GDP growth for the year at 7.7 per cent, slightly higher than the second advance estimate of 7.6 per cent released in February. The print is above the potential growth rate of 7 per cent noted in the recent Economic Survey and much stronger than projected by policymakers and analysts at the beginning of the year.

The revision is not unusual, considering the February estimate was based on incomplete information for the fourth quarter. High-frequency indicators such as auto sales and retail credit growth indicate domestic demand held up during the quarter. But what really makes the performance stand out is a combination of sustained high growth — measured through rebased GDP — and benign inflation, at a lower-than-expected 2.1 per cent.

Several factors helped sustain growth despite adverse external conditions. The impact of high tariffs imposed by the US was lower than initially feared as exporters front-loaded shipments. Services exports remained robust, while exemptions for fast-growing sectors such as electronics helped limit the damage. Low crude oil prices and a normal monsoon — the “good luck” factor for India — supported growth and kept inflation under control.

Policy interventions contributed, too. Rate cuts by the RBI's Monetary Policy Committee supported financial conditions and demand. Fiscal measures, including rationalisation of GST rates, income tax relief and an increase in direct benefit transfers at the state level, provided additional support. These measures helped strengthen household purchasing power and business sentiment. Private consumption and investments emerged as the primary drivers last year, growing at 7.7 per cent and 8.2 per cent respectively.

Cut to this fiscal year.

The ride is becoming more laboured. Growth expectations are being lowered as downside risks to the economy begin to materialise.

Crisil expects GDP growth to slow to 6.6 per cent from 7.7 per cent last year. The RBI, in its June monetary policy, also pared its growth outlook to 6.6 per cent, while keeping rates and stance unchanged. A slew of measures announced by authorities was aimed at encouraging foreign capital inflows, which, along with a depreciating rupee, have been a wrinkle in an otherwise healthy macroeconomic story.

Though the government is likely to maintain its investment momentum, a pick-up in private corporate investment will be delayed due to a highly uncertain environment.

Inflation, on the other hand, will move up to 5.1 per cent, with risks tilted to the upside. Despite the slowdown in real GDP, the nominal GDP growth is set to be higher due to higher inflation based on both the Wholesale Price Index (WPI) and Consumer Price Index (CPI).

The main reason for the worsening growth-inflation mix is the West Asia conflict that has brought



DHARMAKIRTI JOSHI

The expected slowdown in major export destinations, could reduce opportunities for goods exports. Also, the two 'good luck' factors that supported growth and kept inflation benign last year have turned adverse

a major energy shock and a wider trade disruption. It has led to a sharp rise in freight and insurance costs, intensified uncertainty and, coming on top of the unresolved tariff shock of 2025, rendered the external environment more challenging.

The expected slowdown in major export destinations, including the US, Europe and West Asia, will likely reduce opportunities for India's goods exports. Also, the two favourable “good luck” factors that supported growth and kept inflation benign last year have turned adverse.

Crude oil is expected to average \$90-95 per barrel this fiscal, compared with \$70 last fiscal. India's high dependence on imported energy increases its vulnerability to both price increases and supply disruptions. Higher costs reduce availability of energy at affordable prices and weigh on growth, particularly in manufacturing, construction and services such as travel, transport and restaurants.

The forecast of El Niño reducing rainfall to 90 per cent of the long-period average could become another headwind to both growth and inflation. Although the development of the Indian Ocean Dipole can provide some offset, it is a bit early to predict how this will play out. The risks from monsoons remain tilted to the downside.

Crisil's Financial Conditions Index shows the conflict significantly tightened India's financial conditions in March and April through capital outflows, sharp depreciation of the rupee and rising bond yields. Tighter financial conditions can affect borrowing costs, investment decisions and overall confidence.

The upside risks to inflation have started materialising, with the pressure currently more visible in wholesale inflation than retail in-

flation. WPI inflation, which is directly affected by global commodity shocks, came in at 8.3 per cent in April, while CPI inflation remained benign at 3.5 per cent.

We expect CPI inflation to rise to 5.1 per cent as manufacturers pass on higher input costs to consumers. The government has also begun passing through higher crude prices to petrol and diesel, which will have both direct and indirect effects on inflation. A weaker rupee further raises the risk of imported inflation.

Food inflation may also come under pressure due to disrupted agricultural production from heatwaves and expected below-normal rainfall. The household survey of inflation expectations released alongside the June monetary policy report suggests rising inflationary expectations.

On the positive side, Crisil Ratings notes that strong balance sheets, with the median debt-to-equity ratio currently at a low of 0.45, provide a cushion to a majority of medium and large corporates.

However, the policy space to support growth through easy monetary policy has evaporated because of rising inflation risks. Fiscal policy will need to balance several priorities: Supporting MSMEs, managing a higher fertiliser subsidy bill and accelerating efforts to improve energy security and affordability.

With the rising frequency of global shocks and geopolitical reset towards protectionism, this is the moment to activate medium-term growth drivers that are within domestic control. Removing the bottlenecks in the economy, fast-tracking ease of doing business and pushing ahead with remaining reforms will be crucial to revive private capex and realise the export potential of recently signed foreign trade agreements.

The writer is chief economist, Crisil

The Ideas Page

MONDAY, JUNE 8, 2026

Policy is undoing what science built. That's a challenge for Cotton Productivity Mission



FROM PLATE TO PLOUGH

BY ASHOK GULATI, AYUSHI GUPTA AND RITIKA JUNEJA

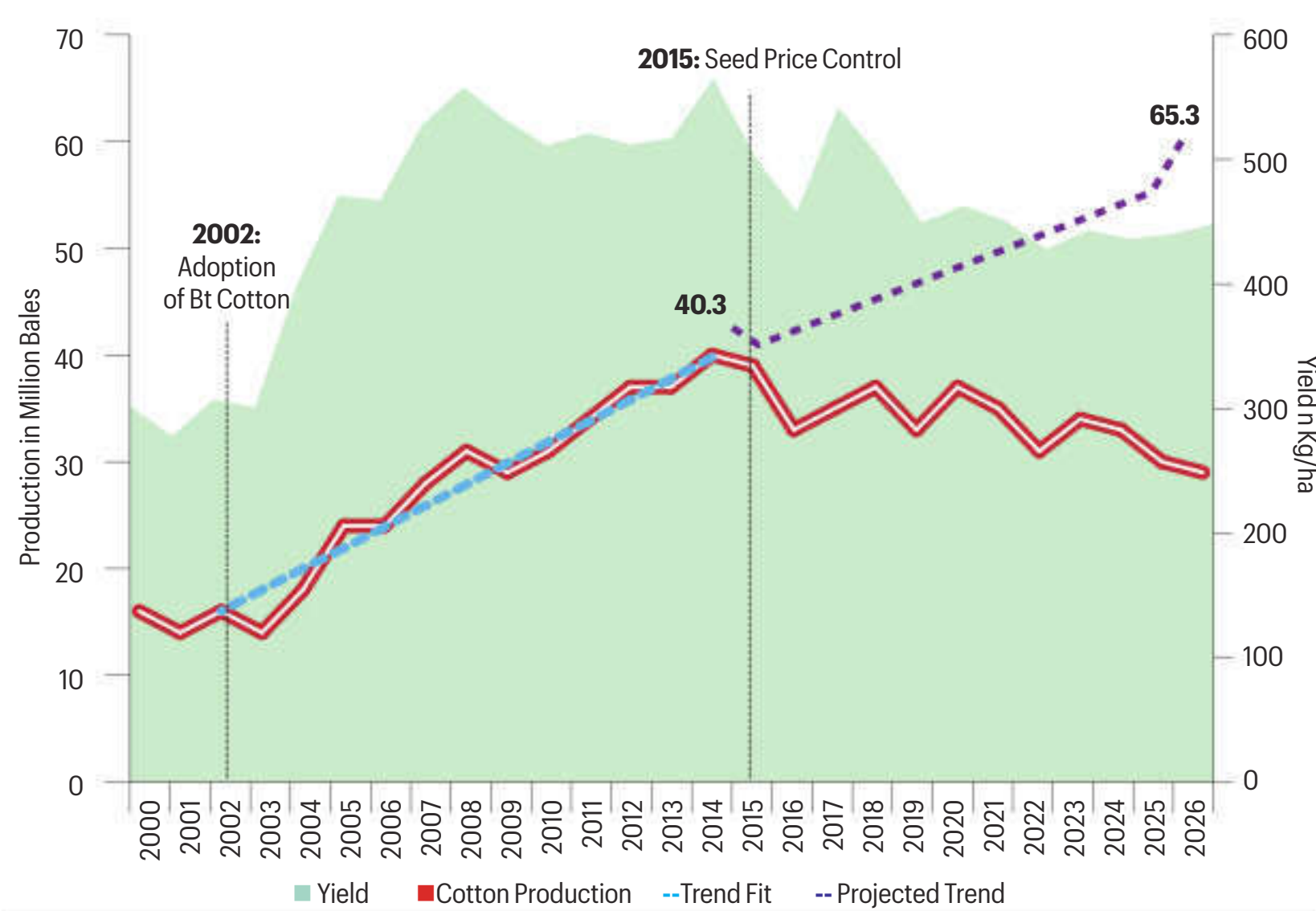
IN APRIL 2026, the Union Cabinet approved the Mission for Cotton Productivity with an outlay of Rs 5,659 crore, to run from 2026-27 to 2030-31. The intent is to raise lint productivity from 441 kg per hectare (in triennium ending (TE) 2025-26) to 755 kg/ha by 2031. But to understand whether this mission can succeed, it is worth recalling where India once stood and how deep the crisis in cotton fields runs.

In 2002, Prime Minister Atal Bihari Vajpayee approved the commercial cultivation of Bt cotton as cleared by the Genetic Engineering Appraisal Committee (GEAC). At the time, he said: "The next big revolution that is unfolding in the world is the biotechnology revolution. This is going to touch the lives of ordinary people in ways that we cannot even fully imagine today. We must not lag behind others in this revolution. India should aspire to be one of the leaders of this revolution. We must plant its healthy saplings in different parts of the country so that we can reap their fruits soon."

India did not lag behind. For a decade, it led. The first Bt hybrids incorporating the cry1Ac gene were approved to resist the bollworm complex. By 2006, GEAC had approved Bollgard II, a second-generation hybrid with two stacked genes. The results were unambiguous. Cotton production rose from 13.6 million bales in 2002-03 to 39.8 million bales in 2013-14, a 193 per cent increase. Annual output grew at 10 per cent, and area expanded by 56 per cent, from 7.6 million hectares (mha) to 11.9 mha. Yields improved by 88 per cent, from 302 kg/ha to 566 kg/ha. India became the largest producer and second-largest exporter of cotton. A technology decision taken by Vajpayee triggered an agricultural revolution. But thereafter, the policies by successive governments, both at the state and central levels, undid what science had built.

After Bollgard II, Mahyco-Monsanto

• PRODUCTION AND YIELD OF COTTON IN INDIA



Biotech (MMB) India developed the next generation: Bollgard II with Roundup Ready Flex, and Bollgard III, incorporating three stacked genes with herbicide-tolerant traits designed to address evolving pest resistance and rising weed-management costs. But they were not released.

What killed them was a sequence of price-control decisions that made seed innovation commercially unviable. In 2006, Andhra Pradesh imposed a Maximum Retail Price of Rs 750 per packet, less than half the prevailing price of Rs 1,600, of which Rs 1,250 had been charged by MMB as trait fee. Maharashtra and Gujarat swiftly matched the cap. In 2015, the Centre formalised this through the Cotton Seed Price Control Order, slashing the trait fee by 74 per cent from Rs 186.95 to Rs 49 per packet. In 2018, the national retail price was cut further to Rs 740. By 2020, the trait fee had been abolished entirely.

The consequence was direct. Faced with unfriendly regulation,

Had India continued on the production trajectory between 2002 and 2014, cotton output in 2026 would have reached 65.3 million bales. The actual figure in 2025-26 is 29 million bales. The gap is the accumulated cost of a decade of policy failure

Monsanto withdrew Bollgard II-RRF from GEAC consideration. Meanwhile, advanced agricultural economies routinely plant Bollgard 3 ThryvOn with XtendFlex, a platform combining three-gene caterpillar protection, built-in biological defence against sucking pests, and tolerance to three separate herbicide groups simultaneously. Brazil, where 92 per cent of cotton is rainfed, has deployed technology and seed innovation to the point where 60 per cent of cotton-planted area supports a second crop within the same agricultural year.

Today, India is not a leader of the biotechnology revolution in agriculture. It's not even a serious participant.

Look at the productivity of cotton: Australia: 2,340 kg/ha; China: 2,311 kg/ha; Brazil: 1,943 kg/ha, and the United States: 976 kg/ha. These are lint yields for the TE 2025-26, reported by the International Cotton Advisory Committee. And India, which has the largest cotton-growing area in the world (11 million hectares), has a yield of just 441 kg/ha.

The gap between India and its competitors is a precise measure of the gap in technology adoption, research investment, and regulatory ambition.

What is painful to see is that since 2014-15, production has declined at an average annual rate of 2 per cent. Had India merely continued on the production trajectory established between 2002 and 2014, cotton output in 2026 would have reached 65.3 million bales. The actual figure in 2025-26 is 29 million bales. The gap, 36.3 million bales, is the accumulated cost of a decade of policy failure. India went from a net cotton exporter to importing 4 million bales in 2025-26. It won't take long for India to become a large importer of cotton if business as usual continues.

It is against this backdrop that one needs to evaluate the Cotton Productivity Mission. The Mission's intent is welcome, but does it have the technology that global giants provide? The simple answer is "no". How will it achieve the productivity target of 755 kg/ha by 2031 remains to be seen. And even if it does, it would still leave India at less than half of what Brazil, China, and Australia have achieved today. How will it compete with them?

Developing new seed varieties requires years of investment in plant breeding, biotechnology, and trait research, followed by a lengthy biosafety regulatory process with no guarantee of commercial approval. Without a policy framework that ensures adequate returns on that investment, private companies have no rational basis for undertaking it. Price controls that compress trait fees to zero do not merely limit profits; they eliminate the incentive to innovate at all.

The government, therefore, faces a choice. Either it restores the intellectual property framework that allows private innovators to recover their costs, which means revisiting the Cotton Seed Price Control Order, or it commits to funding public sector R&D at a scale sufficient to fill the gap the private sector has vacated. What is not sustainable is the current arrangement, where price controls suppress private innovation, public R&D is chronically underfunded, and farmers are left with dated technology. Can the Cotton Productivity Mission undo this policy damage? Only time will tell.

Gulati is distinguished professor, Gupta a research associate, and Juneja a senior fellow at ICRIER. Views are personal

Who gets rupee right: Economists, market wallahs or moralists?



ISHAN BAKSHI

IN TIMES of crisis, advice, often unsolicited, tends to pour in from all quarters. This time is no different. Economists, market *wallahs* and moralists — all have given their two cents on how the ongoing crisis ought to be tackled. What they have proposed, though, has been largely at odds. That should not be surprising. After all, the advice that is given, as is often the case, depends on where you sit.

Economists have, by and large, stuck to orthodoxy — that the central bank should not defend the rupee. It should neither draw down its forex reserves nor should it raise interest rates to defend the currency. An inflation-targeting central bank should focus on inflation, not the exchange rate. The rupee should, instead, be left to find its own level. A weaker currency would, at the very least, help maintain the competitiveness of the country's exports. It would also make Indian assets such as stocks and bonds cheaper in dollar terms, increasing their attractiveness for foreign investors.

Efforts to defend the currency are, in any case, likely to be futile as the rupee's problems run deep. The energy shock that has exposed India's vulnerability, the absence of a domestic AI play alongside the rapid global adoption of AI models that threatens India's growth engine, the possibility of the China+1 dream fading away and the risk-return calculus on domestic investments souring even for Indian firms — all raise deeper questions about the country's growth prospects. It is these concerns that have prompted a flight of capital and put pressure on the currency that need to be addressed, not the rupee's decline.

Market *wallahs*, including some from India Inc, however, have a different view, with many in favour of supporting the currency. Some have even called for raising interest rates to defend the currency. Never mind that the economy isn't overheating — core inflation (excluding precious metals) is around just 2.2 per cent. But, for them, "100" is not just another number. Worries of a vicious cycle — the decline in the rupee triggering more capital outflows, which in turn compounds balance of payments pressures and further weakens the currency — have gained traction. The Indian markets, which find little mention among global investors at the moment, are fearful of a further loss of investor confidence. The defence of the currency is, in some sense, also a defence of the markets.

The Indian central bank has often tended to side with this line of thinking despite its pronouncements to the opposite. There have been numerous

episodes when, in the face of sustained depreciation pressures, it has intervened through various tools in order to defend the rupee — an approach it has pursued across governments of varying ideological dispositions.

But, if there is one thing that both economists and the market *wallahs* agree on, it is the need to facilitate foreign capital flows through various channels. While their preferences may differ, the proposals have ranged from eliminating taxes on investments of foreign institutional investors in markets to encouraging flows through FCNR deposits and/or ECBs, and facilitating greater FDI flows, among others. Some of these suggestions have been acted upon. On Friday, the government and the central bank announced a slew of measures to attract foreign capital. But the question is: In an environment where the India-US yield differential isn't that large, how much can be realistically mopped up? And would it be enough to ease the pressure? If not, will the RBI follow this up with an interest rate defence as the market *wallahs* have argued for?

The moralists, on the other hand, have a different take. In times of distress, personal sacrifices must be made, and those profiting, the speculators, must be punished. So, there is a need for market interventions and restrictions on economic activities, demands for austerity and exhortations to curb the use of foreign exchange.

Of course, the distinction between these categories isn't neat. Overlaps do exist. But all these groups are currently jostling to influence the direction of policy at this critical juncture for the Indian economy. At the moment, it does appear that the market *wallahs* and the moralists have the upper hand. They may well believe that their measures will help tide over this crisis. But any relief is likely to be short-lived. The focus should instead be on addressing the core issues facing the economy.

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URVASHI PRASAD

THE HEADLINE from NFHS-6 on child health is heartening. Stunting — children who are too short for their age — has declined nationally from 35.5 per cent to 29.3 per cent. That is a substantial improvement and deserves recognition. However, read in full, NFHS-6's child nutrition story presents a more complicated picture: National gains are uneven, several large states have deteriorated, and key drivers of child nutrition remain deeply concerning.

Severe wasting, the most dangerous form of acute malnutrition, declined from 7.7 per cent to 5.2 per cent. This is a meaningful gain that should translate into fewer child deaths.

Yet overall wasting fell only marginally, from 19.3 per cent to 19 per cent, meaning nearly one in five Indian children under five remains acutely malnourished. The proportion of underweight children moved from 32.1 per cent to 31.8 per cent. Nearly one in three Indian children is still underweight.

The state-level data is even more troubling. In Jharkhand, underweight prevalence increased from 39.4 per cent to 41.1 per cent. In UP, wasting rose from 17.3 per cent to 19.2 per cent and underweight from 32.1 per cent to 34.5 per cent. Rajasthan recorded similar deterioration, with wasting rising from 16.8 per cent to 19.8 per cent and underweight from 27.6 per cent to 33.3 per cent. These setbacks in some of India's largest states are obscured by the national headline.

The stunting figure itself conceals wide disparities. Kerala's stunting rate is 20.1 per

cent, compared with 35.6 per cent in Bihar and 35 per cent in Jharkhand. The gap reflects two very different childhood realities under the same national programmes. Jharkhand's wasting rate stands at 22.3 per cent, more than double Kerala's 10.9 per cent, and virtually unchanged from NFHS-5's 22.4 per cent despite four years of programme implementation.

Only 15.3 per cent of children aged 6-23

Breastfeeding remains one of the most evidence-backed, low-cost public-health interventions available. Its decline in states with high malnutrition burdens is a serious policy failure

months receive an adequate diet, up from 11 per cent. This is progress, but it still means more than eight in 10 children during the most critical developmental period are not receiving what nutrition science recommends. Dietary adequacy is only 8.7 per cent in Rajasthan and 11.9 per cent in Bihar. The decline in stunting has occurred despite this gap, not because it has been resolved.

Perhaps the most worrying finding is breastfeeding. Exclusive breastfeeding among infants under six months has fallen nationally from 63.7 per cent to 55.8 per cent. In UP, it has dropped from 59.7 per cent to 34.6 per cent. Jharkhand and Rajasthan have also seen sharp declines. Kerala, in contrast, improved from 55.5 per cent to 72.7 per cent. This suggests the problem is not institutional delivery but postnatal support after discharge. Breastfeeding remains one of the

most evidence-backed, low-cost public-health interventions available, benefiting both child survival and maternal health. Its decline in states with high malnutrition burdens is a serious policy failure.

There are clear successes. Full vaccination coverage increased from 83.8 per cent to 87.1 per cent, while rotavirus vaccine coverage surged from 36.4 per cent to 85.4 per cent, reflecting effective national rollout.

NFHS-6 confirms India can reduce stunting. But wasting, dietary inadequacy and declining breastfeeding remain major risks. Whether the next NFHS shows continued progress on stunting will depend on whether these warning signs are addressed now.

The writer is senior fellow, Pahle India Foundation and former director, office of vice chairperson, NITI Aayog

Bangladesh & Turkey agree to strengthen defence, political ties

Press Trust of India
Dhaka, June 7

BANGLADESH AND Turkey have reached an understanding to form a joint ministerial level committee on defence and foreign affairs to strengthen bilateral political cooperation.

The decision was announced Saturday after a meeting between Prime Minister Tarique Rahman and Turkish Foreign Minister Hakan Fidan, who arrived in Dhaka on a three-day visit.

"Bangladesh and Turkiye have decided to form a joint committee at the ministerial level on defence and foreign affairs to strengthen bilateral relations," the state-run BSS news agency reported quoting the prime minister's press wing.

The meeting decided to hold annual foreign office consultations between foreign and defence ministers of the two countries to further deepen the



Turkish Foreign Minister Hakan Fidan. FILE

“strategic relations,” the press wing spokesman said.

Fidan said his visit constituted the first step towards elevating Bangladesh "Turkey relations to the "strategic level" as he simultaneously conveyed President Recep Tayyip Erdogan's greetings to Rahman.

The press wing said the prime minister's discussion with Fidan also focused on the Rohingya crisis, climate change, trade and investment, joint production and cooperation, and regional and international matters of mutual interest.

‘GOVT CAN ISSUE FOREIGN CURRENCY SOVEREIGN BONDS IF MORE STEPS NEEDED’

‘Forex measures, FII bond tax leeway could attract \$60-80 bn’

DEPENDING ON how they are executed and the final guidelines, India could see inflows to the tune of \$60 billion-\$80 billion from the various measures announced by the government and the Reserve Bank of India (RBI) on Friday to attract foreign capital — from the scrapping of capital gains and withholding tax on foreign institutional investors’ (FIIs) investment in government debt to concessional windows to boost foreign currency deposits and loans — according to Devang Shah, Head of Fixed Income at Axis Mutual Fund. And while these steps appear adequate, there remains space to take more action, if needed.

“Issuance of foreign currency sovereign bonds could serve as a supplementary step to finance the fiscal deficit, ease pressure on domestic yields, and support foreign inflows,” **Devang Shah** told **Siddharth Upasani** in an interview. Edited excerpts:

What is your estimate of the foreign inflows that we can see from the measures announced by the government and the RBI?

There were two sets of measures. The government removed capital gains and withholding taxes on government bonds. This itself is a positive measure which would attract flows and is expected to serve as a precursor to government bonds being included in Bloomberg’s indices. If that materialises, it could potentially result in additional inflows of

\$20 billion-\$25 billion.

Significant importance must also be attached to the second set of measures announced by the RBI such as the Foreign Currency Non-Resident (Bank) deposits, for which the entire hedging cost will be provided. In addition, there is the concessional forex swap facility for External Commercial Borrowings (ECBs) by Public Sector Undertakings (PSUs). While the final guidelines are awaited, it may be understood that the RBI would have received feedback from market participants for the same. If the concessional rates are sufficiently attractive, ECB borrowings by PSU lenders could become a significant avenue.

The combined impact of these two facilities could possibly bring in inflows of \$50 billion-\$55 billion, is our estimate.

Further, we believe that the removal of capital gains and withholding taxes is expected to deliver meaningful benefits over time. Yields in India have already increased notably, while the rupee has depreciated by over 15% in the past 18 months. At these levels, certain FIIs and hedge funds may find yields attractive. To summarise: \$30 billion-\$35 billion may come from FCNR(B) deposits, \$10 billion-\$20 billion from PSU ECBs, and \$20 billion-\$25 billion from the tax concessions and Bloomberg index inclusion.

In aggregate, this could translate into inflows of \$60 billion-\$70 billion over the next 12-18 months.



DEVANG SHAH
HEAD - FIXED INCOME
AXIS MUTUAL FUND

Are the withholding and capital gains taxes the only reason why FII holdings in Indian government debt have been so low? Or does more need to be done?

Several measures have been announced, and their effectiveness will depend on how they are executed before any additional steps are considered. However, if these measures do not prove as attractive as anticipated for any reason, and if the West Asia crisis remains unresolved, some pressure may emerge on the fiscal front.

At present, even after the recent increases in petrol and diesel prices, the Centre could face a shortfall of Rs 2.5 lakh crore-Rs 3 lakh crore. This gap may widen further if crude oil prices rise beyond \$100-\$110 per barrel. In such a scenario, the government could consider issuing foreign currency sovereign bonds.

While the measures announced so far appear adequate and could attract inflows in the range of \$60 billion-\$80 billion,

depending on execution and final guidelines, there remains room for additional action if required. Issuance of foreign currency sovereign bonds could serve as a supplementary step to finance the fiscal deficit, ease pressure on domestic yields, and support foreign inflows.

Could there be a negative impact if the numbers fall short of expectations?

Even if the eventual inflows are lower than current estimates, this would still represent a positive step. The measures announced were not expected to be introduced all at once, but rather over a period of time, which underscores the decisiveness of both the government and the RBI. We believe that the measures were announced to acknowledge the underlying risk of a potential Balance of Payments (BoP) deficit. Based on the market estimates, the BoP gap for the year could be in the range of \$50 billion-\$75 billion, and the measures collectively aim to address this.

There is also limited concern around execution. A relevant precedent exists from 2013, when banks successfully mobilised approximately \$26 billion through FCNR(B) deposits with effective efforts. In that context, achieving inflows of \$30 billion-\$40 billion under the current framework appears realistic.

Do you expect FIIs to pour money into Indian debt immediately or wait for more cues on the interest

rate front?

There are broadly two types of investors: those with a medium-term outlook and others with a more tactical approach to interest rates. From a macroeconomic standpoint, we believe that the RBI has attempted to prepare the market by highlighting elevated inflation risks and providing clear forward guidance. It has indicated that if inflation data deteriorates further, policy rate hikes remain a possibility.

Macroeconomic data will ultimately determine whether FIIs choose to take positions immediately or adopt a wait-and-watch approach. However, initial indications suggest healthy participation given that FII activity in the debt market on Friday was robust, with estimated inflows in the range of Rs 3,000 crore-Rs 4,000 crore based on changes in holdings. That being said, a segment of investors may prefer to wait for greater clarity on the interest rate cycle. It is important to note that rates in India have already moved up meaningfully. The 10-year term gov bond yield, which had bottomed at 6.15% in June 2025, had risen to around 7% prior to the latest announcements. Historically, in rate upcycles over the past decade, 10-year gov bond yields have not moved significantly beyond 7.5%. Long-term gov bond yields are already in the range of 7.6-7.65%, indicating that a substantial portion of the adjustment has already occurred.

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INSIDE ANDHRA’S AI AMBITIONS

Why Andhra Pradesh plans to let data centres buy & distribute own electricity

Pratyush Deep
New Delhi, June 7

DATA CENTRES, the massive facilities that power artificial intelligence (AI), are among the largest consumers of electricity in an economy. As the AI boom drives up the demand for computing power, governments are increasingly being forced to rethink how these facilities access and procure electricity.

The Andhra Pradesh government, which has lofty AI ambitions for the state, has announced a first-of-its-kind policy move to address this issue — let data centres function under the tag of a power distributor by granting them a deemed distribution licence (DDL).

DDLs are not a new concept in India’s electricity framework. They have largely been issued to government entities, special economic zones, ports, airports and industrial enclaves. Even the Railways, until recently, was a deemed distribution licensee for around a decade. Extending such a licence to private data centre developers is a major departure from the norm.

But what exactly are these licences and how will it help data centres? Very simply put, it will allow them to procure electricity at a relatively lower cost and distribute it within their own campuses. But there’s more to this. Here’s a deeper dive into the mechanics of how this policy will work.

Data centres’ massive electricity needs

AI runs on algorithms, but those algorithms require vast computing power housed in data centres. Data centres, in turn, run on massive quantities of electricity. This is why data centres are identified not by the number of servers they house, but by the amount of power they require. As AI adoption accelerates, the demand for computing capacity is driving a surge in data centre construction — and with it, a sharp rise in electricity requirement. In the US, home to more than half of the world’s data centres by both numbers and power consumption, the rapid expansion of hyperscalers has emerged as a political and regulatory issue, with concerns that growing power demand from data centres is contributing to higher electricity prices in regions where data centres are lo-

• POWERING GROWTH

ANDHRA PRADESH is leading the race to become a major data centre hub among Indian states

GOOGLE IS setting up a 1 GW data centre campus in Visakhapatnam in one of the largest such projects in the country

THE STATE is also planning to create 5 GW of data-centre capacity in Visakhapatnam alone

THE COUNTRY’S installed data-centre capacity currently stands at around 1.2 gigawatt



but is estimated to quadruple by 2030

cated. Such concerns could surface in India as well. The country’s installed data-centre capacity currently stands at around 1.2 GW but is estimated to quadruple by 2030. Among Indian states, Andhra Pradesh is leading the race to become a major data centre hub.

Google is setting up a 1 GW data centre campus in Visakhapatnam in one of the largest such projects in the country. And the state is planning to create 5 GW of data-centre capacity in Visakhapatnam alone.

How will DDLs help?

Unlike the US, electricity tariffs for industrial and commercial consumers in India are substantially higher than those for residential and agricultural users. This is largely because industrial and commercial consumers essentially subsidise agricultural and domestic consumers. To understand why DDL status matters for data centres, it is important to understand how electricity tariffs are structured.

Broadly, a consumer’s electricity bill has two components: fixed charges and energy charges. Fixed charges have to be paid irrespective of how much electricity is consumed, while energy charges vary with actual consumption. The fixed-cost component covers expenses such as capacity payments made to power generators, transmission infrastructure costs, salaries of utility employees, and maintenance of the electricity network. When a distribution company (discom) acquires a large new consumer, it must ensure adequate power

supply for that consumer by contracting additional generation capacity and, where necessary, expanding transmission and distribution infrastructure. These costs eventually find their way into consumer tariffs. This becomes particularly relevant in the case of data centres, whose electricity demand can run into hundreds of megawatts — comparable to that of a medium-sized city. Andhra Pradesh’s policy allows data centres with a minimum connected load of 300 MW to obtain DDL status, with investors permitted to aggregate loads from multiple facilities to meet the threshold.

By granting DDL status, the state effectively allows these large consumers to arrange their own power procurement and develop electricity distribution infrastructure within their project boundaries. As a result, the local discom is not required to make fresh generation capac-

ity commitments or undertake network investments specifically to serve these energy-intensive facilities. For data centre operators, the benefits could be substantial. DDL holders can procure power from any lawful source, including through open-access arrangements, while being exempted from the cross-subsidy surcharge and additional surcharge that are typically levied on open-access consumers. This allows them to avoid the higher tariffs that commercial consumers usually pay to subsidise residential and agricultural users.

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{ WEEKLY CAPSULE }

New diabetes, obesity pill

Scientists at Karolinska Institutet have developed an experimental diabetes and obesity pill that works in a completely different way from drugs like Ozempic. Rather than reducing hunger, it activates metabolism in skeletal muscle, helping lower blood sugar and increase fat burning while preserving muscle mass. Early clinical results suggest the treatment is safe and well tolerated. Researchers say this approach improved blood sugar regulation and body composition in animal studies while avoiding several side effects linked to GLP-1 therapies. The treatment is taken as a tablet rather than an injection.



Organ that determines longevity

A long-overlooked organ may hold surprising clues to healthy ageing and cancer survival. Researchers at Mass General Brigham used AI to analyse CT scans from tens of thousands of adults and found that people with healthier thymuses — a small immune-system organ once thought to become largely irrelevant after childhood — lived longer and had substantially lower risks of heart disease, cancer and death. The same organ was also linked to better responses to cancer immunotherapy. Located in the chest, the thymus helps train T cells, a type of immune cells that shield the body against infections and disease.



CKD now a silent health threat

As the global population has grown and aged, chronic kidney disease (CKD) has moved into the top 10 causes of death worldwide for the first time. A global analysis found that the number of people living with the condition rose from 378 million in 1990 to 788 million in 2023. The work was led by researchers at NYU Langone Health, the University of Glasgow, and the Institute for Health Metrics and Evaluation (IHME) at the University of Washington. Impaired kidney function was also a major risk factor for heart disease, contributing to about 12 per cent of global cardiovascular deaths.



Oversight is a key reason why fatty liver disease is undiagnosed

Why fatty liver warning is easy to ignore

Consult a doctor when bloating is persistent or occurs without a trigger

A 42-YEAR-OLD corporate professional had been experiencing frequent bloating for several months. She assumed it was due to irregular meals, work-related stress and occasional overeating. Antacids provided temporary relief but the discomfort kept returning. During a routine health check-up, an abdominal ultrasound revealed fatty liver disease. Apart from being overweight and having borderline high cholesterol, she had no obvious symptoms. Many like her often experience bloating after a heavy meal, eating too quickly or consuming certain foods. In many cases, it is linked to common digestive issues such as acidity, indigestion or gas. However, when bloating becomes frequent, persistent or occurs without an obvious trigger, it may be worth evaluating further. It could be indicative of fatty liver disease, which is becoming increasingly common among Indians.

While bloating alone does not indicate a liver problem, recurrent abdominal discomfort can sometimes be a subtle sign that something more may be going on beneath the surface. The liver plays a vital role in metabolism, digestion, and nutrient absorption. As fat deposits increase within the liver, the organ can become enlarged. While this may not cause significant pain, some individuals may experience a sensation of fullness, heaviness, or mild bloating in the abdomen. If this occurs regularly, persists for weeks or develops without context, it may be worth investigating further.

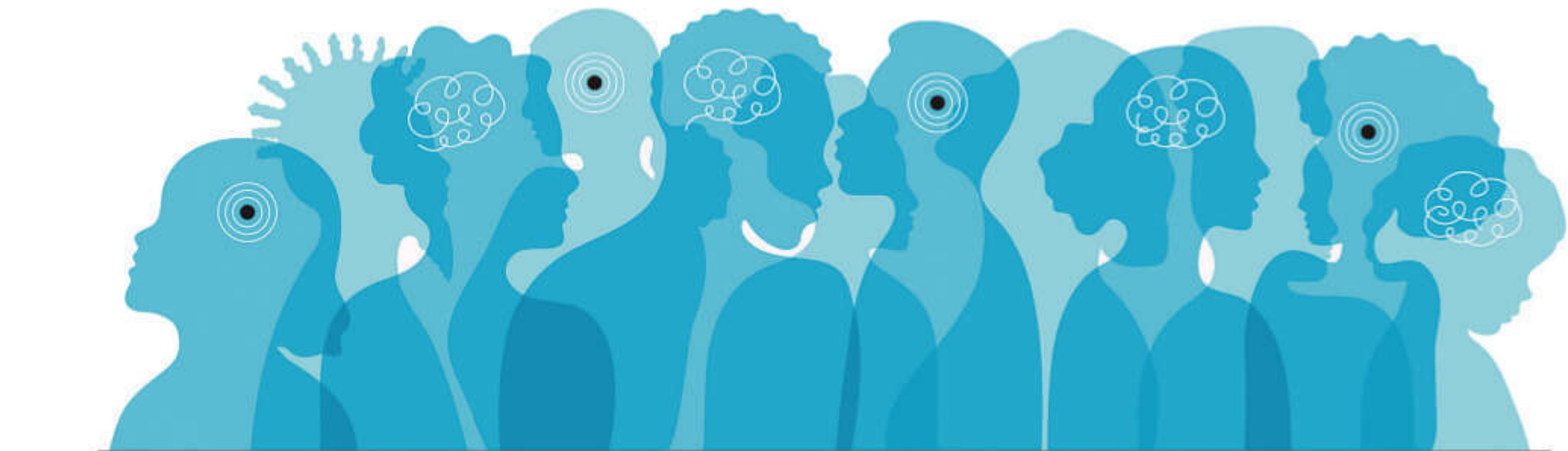
As fat deposits increase within the liver, the organ can become enlarged. This triggers fullness

Oversight is one of the key reasons why fatty liver disease frequently remains undetected until it is identified during a routine health check-up or imaging test performed for another reason. A major challenge with fatty liver disease is that many people do not experience any symptoms at all. The condition is often discovered incidentally during routine blood tests or abdominal ultrasounds. This makes awareness and regular health screening particularly important for individuals who are overweight or living with diabetes, high cholesterol or high blood pressure. Some individuals may experience non-specific symptoms such as fatigue, reduced energy levels, a feeling of fullness in the upper right side of the abdomen, or difficulty losing weight. Because these symptoms are subtle and can be attributed to several other conditions, they are frequently overlooked.

The increasing prevalence of fatty liver disease reflects broader lifestyle and metabolic health trends. Obesity, Type 2 diabetes, hypertension and abnormal cholesterol levels are becoming increasingly common across India. Contributing factors include physical inactivity, excessive consumption of calorie-dense foods, irregular meal timings, chronic stress, and inadequate sleep.

What makes the situation more concerning is that fatty liver is no longer a condition seen only among older adults. Increasingly, it is being diagnosed in younger individuals, many of whom are asymptomatic. This underscores the importance of paying attention to seemingly minor health changes, particularly in people with underlying metabolic risk factors. Simple investigations such as liver function tests, blood sugar assessments, lipid profile testing, and abdominal ultrasound scans can often provide valuable insights into the source of persistent symptoms.

If left unmanaged, fatty liver disease can progress beyond simple fat accumulation. In some individuals, it may lead to liver inflammation, a condition known as steatohepatitis, which can eventually progress to fibrosis, cirrhosis and even liver failure. The encouraging news is that fatty liver disease is often reversible in its early stages. Managing body weight, engaging in regular physical activity, following a balanced diet, improving sleep quality and maintaining healthy blood glucose and cholesterol levels can reduce complications.



GETTYIMAGES

Rinku Ghosh

At 46, Amit Mehra* is doing what he loves most — playing cricket, travelling with his family and maintaining a busy professional life as a chartered accountant. Looking at him today, it is difficult to imagine that eight years ago he was diagnosed with a tumour in one of the most critical regions of his brain. “At 38, my whole world had come crashing down. Now I look forward to my weekend cricket match and know when to log out of work,” he says.

Mehra is one of the young Indians reporting brain and Central Nervous System (CNS) tumours in India which have an estimated incidence of 30 cases per 100,000 population. Roughly 30% to 35% of these tumours are malignant (cancerous), while the majority are benign (non-cancerous).

Mehra began noticing brief episodes of weakness and numbness in his left hand while playing cricket. Sometimes he would struggle to hold on to objects. The episodes lasted only a few seconds before disappearing. He attributed them to physical strain, poor posture or fatigue. “It felt like numbness from sleeping awkwardly or overexertion,” he recalls. “The symptoms were so generic that I never imagined they could be related to brain tumour.” Mehra was worried because he kept to a good fitness discipline, ate clean, had no diabetes or hypertension and wasn’t a smoker or alcohol drinker.

That’s the reason he paid attention to his symptoms, which though fleeting in nature, were replicative. An MRI scan revealed a tumour located in the brain’s eloquent cortex — a region responsible for critical functions such as movement and sensation. What appeared to be transient weakness was actually focal motor seizure triggered by the tumour. A surgery and eight years later, there has been no recurrence. His medications have been significantly reduced and he remains active. “I just got a promotion at my firm for high productivity,” he says. His story reflects a larger shift in brain tumour care. Increasingly, a diagnosis that once carried an almost universally grim prognosis is becoming a condition that can often be treated successfully, with many patients returning to work, family life and even sports.

More cases or better detection?

Brain tumours are being diagnosed more frequently in India than ever before. While the incidence remains relatively low compared to other diseases, neurologists say awareness and access to advanced imaging have transformed diagnosis. “Brain tumours are not necessarily increasing dramatically because of new environmental exposure,”

Brain tumour is not a life sentence

How advances in surgery, molecular diagnostics and multidisciplinary care are helping young patients live longer

says Dr Kapil Singhal, director, Neurology, Medanta, formerly of AIIMS, Delhi. “What has changed significantly is our ability to detect them. MRI scans are more accessible and people are getting evaluated earlier.”

He acknowledges seeing more young and middle-aged adults with brain tumours. Unlike diabetes or heart disease, most brain tumours do not have clearly identifiable lifestyle-related causes. The uncertainty surrounding why they occur often makes the diagnosis unsettling for younger patients who otherwise consider themselves healthy.

Understanding brain tumours

A brain tumour occurs when cells within the brain begin dividing uncontrollably. “It develops when cells in the brain or surrounding tissues acquire genetic mutations that disrupt their normal growth controls. These mutations alter the instructions encoded in DNA, causing cells to multiply and survive longer than they should. As these abnormal cells accumulate, they can form a mass or tumour,” says Dr Singhal.

In most cases, the exact reason behind these genetic changes remains unknown. While some mutations arise spontaneously during a person’s lifetime, others may be inherited from a parent, increasing the risk of developing certain types of brain tumours. “And we need to bust the myth that routine mobile phone usage causes brain tumours. There is no scientific evidence. Also, most brain tumours are not hereditary. They can happen due to random cellular mutations,” explains Dr Singhal.

Contrary to popular belief, not all brain tumours are cancerous. “Many are benign or low-grade lesions that grow slowly and can be completely removed. Modern surgical methods have made longevity and quality of life possible. Low-grade tumours (Grade 1 and Grade 2), which tend to grow slowly have lower recurrence rates. High-grade or malignant tumours (Grade 3 and Grade 4) are more aggressive, have high recurrence chances and require surgery, radiation and



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chemotherapy,” adds Dr Singhal.

Mehra’s tumour was classified as a Grade 2 lesion — a category associated with relatively favourable outcomes when detected early and removed effectively.

“The words ‘brain tumour’ immediately make people think of death or severe disability. But that is simply not true for a significant proportion of patients. Many tumours are treatable, and some can be cured completely,” says Dr Singhal. Apart from primary brain tumours, there are metastatic tumours, too, which spread to the brain from cancers elsewhere in the body.

A revolution in the OR

Perhaps the biggest change over the past decade has been technological. Modern neurosurgery is being driven by sophisticated imaging, computer-assisted planning and real-time monitoring. Before surgery, doctors perform brain mapping using advanced MRI techniques and tractography, which identifies critical nerve pathways responsible for movement, speech and sensation.

“We perform the surgery virtually before doing it real-time,” explains Dr Manish Vaish, director of neurosurgery, Medanta. The imaging data is fed into specialised navigation systems that allow surgeons to plan the safest possible route to the tumour. “The tumour’s location in Mehra’s case made surgery particularly challenging. It was situated in the eloquent cortex, the part of the brain that governs movement and sensation. Even a few millimetres of error could potentially result in permanent neurological impairment,” says Dr Vaish. Inside the operating theatre, multiple technologies work together. Special dyes can be administered that are absorbed preferentially by tumour tissue, causing the tumour to fluoresce under specialised microscopes. Four-dimensional ultrasound helps distinguish tumour tissue from normal brain tissue in real time. Electrophysiological monitoring continuously assesses the function of critical brain regions during surgery. In selected patients like Mehra, surgeons may even perform



an awake craniotomy, allowing communication with the patient during the procedure. “We engaged Mehra by talking about cricket and its statistics. Anything incoherent is a red flag,” recalls Dr Vaish.

He also describes how advanced ultrasonic aspirators have made a delicate surgery procedure easier over the last decade or so. “In simple terms, this surgical tool has a tip that vibrates at very high frequencies (ultrasound), which breaks soft tumour tissue into tiny fragments. At the same time, the device gently suctions away the broken pieces. This is especially useful in brain surgery because tumours are often located close to critical structures that control movement, speech, vision, or memory,” says Dr Vaish. Depending on the complexity of the case, surgeries may last anywhere from three hours to more than 12 hours.

Changing outcomes

One of the most significant advances in neuro-oncology has come not from the operating room but from the laboratory. In the past, doctors relied on what tumours looked like under a microscope. Today, molecular profiling and immunohistochemical (IHC) analysis provide a far deeper understanding of tumour behaviour.

Following surgery, tumour samples are analysed for specific molecular markers and genetic alterations that help doctors predict aggressiveness, recurrence risk and likely response to treatment. “These molecular signatures have become extremely important,” says Dr Vaish. “They help us personalise treatment and provide a much more accurate understanding of prognosis.”

Modern outcomes are no longer dependent on surgery alone. Today’s patients benefit from integrated teams of specialised anaesthetists, critical care specialists, radiation oncologists, rehabilitation experts, nutritionists, psychiatrists and nursing teams.

Back to the mainstream

Recovery timelines vary according to tumour location and complexity. Many patients begin walking within days. Those with low-grade tumours may return to work within weeks. “I went home in three days, worked out of home as I was immune-compromised, walked indoors, reduced screen time to avoid stressing my brain. In three months, I was back at the office and in six months, I was driving again,” says Mehra. Even patients requiring radiation therapy for malignancy resume normal activity during treatment.

Mehra never misses his MRI scans every six months. “That alone has helped me regain something that many patients fear losing after a diagnosis: Normalcy.”

(NAME CHANGED TO PROTECT PRIVACY)

• DECODING A STUDY

Why Kerala and Bengal face Nipah with varied outbreak risks

Study reveals how ecology, human behaviour and preparedness shape Nipah spillovers

Anuradha Mascarenhas

IN JANUARY, two healthcare workers at a private hospital in Barasat, West Bengal, tested positive for Nipah virus. Investigators traced the outbreak to a 55-year-old woman who had consumed raw date palm sap purchased from a local vendor — a known food-borne spillover route also seen in Bangladesh. She later died of acute respiratory distress syndrome and multi-organ failure.

Both healthcare workers had direct, unprotected exposure to the patient — one during routine nursing care and the other while performing cardiopulmonary resuscitation (CPR). Genomic sequencing showed a 99% match with the Bangladesh genotype 2022 strains, confirming hospital-acquired transmission. The outbreak highlights a key

finding from a recent *Nature* report, “Two geographies, one virus: What recurrent Nipah spillover in India reveals.” It says that although Nipah virus circulates in the same bat reservoir, outbreaks in India follow two distinct patterns in West Bengal and Kerala.

“To understand Nipah virus in India, it is important to look beyond virology and consider ecological factors and the strength of health systems in shaping outbreak outcomes,” said Dr Pragya Yadav, corresponding author of the report and director-in-charge of the National Institute of One Health, Nagpur. According to her, Kerala experiences more frequent spillover events but outbreaks are usually contained to single cases or small clusters because of strong surveillance and a One Health approach. In contrast, West Bengal has seen fewer spill-



Attempts to reduce risk by disturbing bat populations or removing roosting trees can sometimes worsen the problem

overs, but these have historically been associated with healthcare-related transmission. Enhanced surveillance was critical in detecting the recent 2026 outbreak.

“The examples of Kerala and West Bengal show that the same virus, maintained in a common reservoir, can produce very different epidemiological outcomes depending on local ecological conditions, human behaviour and health-system preparedness,” say study authors Deepak Y. Patil, Rima R.

Sahay, Sreelekshmy Mohandas and Dr Yadav. But ecological and environmental factors play a stronger, influencing role, they explain.

In Kerala, Nipah outbreaks have consistently occurred between April and September, when hot, humid conditions coincide with fruiting seasons. During this period, bats feed near human settlements, increasing the likelihood of contamination of fruits and surfaces. Bat colonies are also commonly found within residential compounds across Kerala, creating opportunities for direct and indirect human exposure. In Bengal, bat roosts are typically located in peri-urban areas, and spillovers are more closely linked to the harvesting and consumption of fresh or fermented date palm sap. The risk peaks during the sap-harvesting season between November and April. The report cautions that attempts to reduce risk by disturbing bat populations or removing roosting trees can sometimes worsen the problem by dispersing bats and expanding the geographic range of exposure.

New Delhi, June 7

But no matter how un-



Alexander Zverev defeated Flavio Cobolli 6-1, 4-6, 6-4, 6-7(4), 6-1 to win his maiden Grand Slam title on Sunday. REUTERS

Equally, it could spur him on to break the Sinner-Alcaraz duopoly that has been established in men's tennis today. Through it all, expect the applause to only be polite and begrudging.